

# Joyce Glass

Max Planck Institute for Astronomy – Königstuhl 17, 69117 Heidelberg

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## EDUCATION

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**M.Sc. in Physics, University of Heidelberg**

Oct 2022 - Summer 2025  
Heidelberg, Germany

Supervisors: Dr. Eduardo Bañados, Dr. Yanu Khusanova

Thesis title: Unveiling star formation at  $z \sim 1.5$  through Paschen- $\alpha$  line emitters with JWST NIRCам

(Preliminary) Grade average: 1.7 (max: 1.0, min: 5.0)

**B.Sc. in Physics, University of Heidelberg**

Oct 2018 - Oct 2022  
Heidelberg, Germany

Supervisor: Prof. Dr. Christian Enss

Thesis title: Evaluating a novel microstructured, superconducting resonator for the analysis of dielectric properties of N-BK7 at 1 GHz and low temperatures ([PDF](#))

Thesis Grade: 1.0, Grade average: 2.4 (max: 1.0, min: 5.0)

## Research Experience

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**Master Thesis, Max Planck Institute for Astronomy**

June 2024 - June 2025  
Heidelberg, Germany

Supervisors: Dr. Eduardo Bañados, Dr. Yanu Khusanova

Topic: Unveiling star formation at  $z \sim 1.5$  through Paschen- $\alpha$  line emitters with JWST NIRCам

Data: JWST NIRCам in the narrow band F470N and a number of broad bands

- Performing forced photometry on narrow and broad band images to identify line emitters
- Fit results from forced photometry to Spectral Energy Distributions with EAZY-PY to constrain sources to  $z \sim 1.5$
- Determining the Paschen- $\alpha$  Luminosity Function
- Estimating the SFRD at  $z \sim 1.5$  and putting it into context with the evolution of galaxies over cosmic time

**Research Internship, Max Planck Institute for Astronomy**

Febr 2024 - May 2024  
Heidelberg, Germany

Supervisors: Dr. Matthias Samland, Dr. Elisabeth Matthews

Topic: Searching for exoplanets with JWST NIRCам imaging of Orion ([PDF](#))

Data: JWST NIRCам in filters F480M and F210M

- Using WEBBPSF to model and subtract PSFs from stars and identify interesting candidates in the glare
- Determining the flux and uncertainties of the star and exoplanet candidates to determine their colors
- Using isochrone models to evaluate the nature of the candidates
- Oral presentation of my work at MPIA in the APEX department and the JWST group

**Summer School**, *Institute of Space Sciences, Barcelona*

July 2024  
*Barcelona, Spain*

Topic: Multiwavelength Approach to Exoplanetary Systems

Lectures on: Exoplanet properties, their relation to host stars, and various detection and characterization methods

## Teaching Experience

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**Tutor for the Physics Beginners' Lab Course**, *Heidelberg University*

Winter 2023/24 - present

Experiments: Fourier Optics (two days experiment), Coupled Pendulum, Viscous Fluids

Tutoring responsibilities:

- Test groups of four to eight students on the required knowledge prior to the experiment
- Explain software needed for the measurements
- Supervise and grade the execution and analysis of the experiment

## Skills

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**Languages:** German (C2), English (C2, Cambridge Certificate), Greek (B1), French (B1), Japanese (A2)

**Programming Languages:** PYTHON (very good)

**OS:** Windows, Mac, Linux

**Other:** Usage of BASH,  $\text{\LaTeX}$ , Microsoft Office

## Voluntary Work

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**Ultimate Frisbee Coach for the University Team**

2023 - present  
*Heidelberg, Germany*

- Teaching newcomers the techniques, rules and spirit of the game
- Help organize yearly participation in the German University Cup
- Organisation of tournaments over the year with other local teams